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SAFETY-RULES DEFINITION

1.Introduction

The Safety Rules Engine serves as the core reasoning component of the system, responsible for transforming detection and matching outputs into meaningful safety decisions. Unlike conventional approaches that rely on single-frame evaluation, this system introduces a **stateful and feature-rich rule-based framework** that evaluates worker safety over time.

The design emphasizes **robustness, explainability, and real-world reliability**, ensuring that safety decisions are not affected by temporary detection failures or environmental noise.

The implemented system integrates multiple engineered features, including temporal memory, negative evidence tracking, visibility awareness, and risk scoring, to achieve accurate and stable classification of worker safety states.

2.Core Safety Features

The Safety Rules Engine is built upon several key features that enable advanced decision-making:

2.1 Stateful Worker Memory

Each worker is associated with a persistent memory structure that stores historical information across frames.

Tracked attributes include:

- Helmet detection history
- Vest detection history
- Missing frame counters
- Missing duration (in seconds)
- Unsafe and uncertain durations
- Negative evidence durations (e.g., NO-Hardhat)

This feature allows the system to perform **temporal reasoning**, ensuring decisions are based on accumulated evidence rather than isolated observations.

2.2 Temporal Memory and Decay Mechanism

The system incorporates a **time-based memory decay mechanism** to manage detection persistence.

- PPE detections are retained for a short duration
- Memory gradually decays over time
- Old observations are discarded automatically

This ensures that:

- Temporary occlusions do not cause false violations
- Outdated information does not affect decisions

2.3 Negative Evidence Tracking

A unique feature of the system is the use of **negative evidence classes**, such as:

- NO-Hardhat
- NO-Safety Vest

Instead of relying only on absence of detection, the system explicitly tracks violation evidence over time.

- Negative evidence is accumulated
- Duration thresholds determine severity
- Strong negative evidence can trigger immediate unsafe classification

This significantly improves the reliability of violation detection.

2.4 Visibility-Aware Reasoning

The system evaluates whether key body regions are visible before making decisions:

- Head region → Helmet verification
- Torso region → Vest verification

If regions are not clearly visible:

- The system avoids unsafe classification
- Worker is marked as **uncertain**

This prevents incorrect decisions caused by occlusion or poor detection conditions.

2.5 Violation Duration Thresholds

To avoid false alarms, the system introduces **time-based thresholds** for violations.

- PPE must be missing for a minimum duration
- Short-term absence is ignored
- Sustained absence triggers unsafe classification

Example:

- Missing helmet for a few frames → ignored
- Missing helmet for several seconds → unsafe

2.6 Multi-State Classification System

Workers are classified into three distinct states:

- **Safe** → Required PPE present or recently observed
- **Unsafe** → Strong or sustained violation detected
- **Uncertain** → Insufficient or conflicting evidence

This classification allows the system to handle ambiguity effectively instead of forcing incorrect decisions.

2.7 Conflict Detection Handling

The system detects conflicting evidence scenarios such as:

- Helmet and NO-Hardhat detected simultaneously
- Vest and NO-Safety Vest detected simultaneously

In such cases:

- Worker is classified as **uncertain**
- Conflict reason is recorded

This ensures the system remains logically consistent.

2.8 Risk Scoring Mechanism

A dynamic risk scoring system is used to quantify safety severity.

The score is calculated based on:

- Worker state (safe, unsafe, uncertain)
- Duration of unsafe state
- Duration of uncertain state
- Negative evidence strength
- PPE missing duration

The final score is mapped to:

- Low risk
- Medium risk

- High risk

This feature enables prioritization of critical safety violations.

3. Alert Generation Features

The system includes a structured alert generation mechanism that converts worker states into actionable outputs.

3.1 Alert Levels

- **Info** → Worker is compliant
- **Warning** → Worker requires review
- **Critical** → Worker is unsafe

3.2 Context-Aware Messaging

Alerts are enriched with:

- Specific violation reasons
- Worker identification
- Supporting notes

Examples:

- “Worker 3 is unsafe: NO-Hardhat detected.”
- “Worker 5 requires review: Head not visible.”

This ensures alerts are **interpretable and actionable**.

4. Summary

The Safety Rules Engine integrates multiple advanced features to ensure accurate and reliable safety evaluation. By combining temporal memory, negative evidence tracking, visibility awareness, and risk scoring, the system achieves a level of reasoning that closely resembles human decision-making.

These features collectively enable the system to operate effectively in complex real-world environments, making it suitable for deployment in industrial safety monitoring applications.

5.Appendix

Image1:Detection and analysis

```
--------------------------------------------------------------------------------
PROCESSING COMPLETED
--------------------------------------------------------------------------------
Frames processed       : 1111
Unique worker IDs      : 61
Total alerts generated: 9803
Elapsed time           : 0:12:02.802708
--------------------------------------------------------------------------------
LATEST FRAME STATE SUMMARY
Safe: 8 | Unsafe: 0 | Uncertain: 0
Info alerts: 8 | Warning alerts: 0 | Critical alerts: 0
--------------------------------------------------------------------------------
SECOND-LEVEL REVIEW SUMMARY
Likely compliant: 44 | Manual review required: 0 | Sustained likely violation: 0 | Insufficient observation: 1
--------------------------------------------------------------------------------
ANALYTICS SUMMARY
Average persons detected       : 9.69
Average helmet matched         : 8.91
Average vest matched           : 7.91
Average full PPE               : 7.33
Helmet compliance rate         : 91.95%
Vest compliance rate           : 81.67%
Full PPE compliance rate       : 75.65%
--------------------------------------------------------------------------------
OUTPUT FILES
Annotated video       : G:\site-safety-vision\outputs\predictions\test1_annotated.mp4
Inference JSON        : G:\site-safety-vision\outputs\predictions\test1_results.json
Review JSON           : G:\site-safety-vision\outputs\predictions\test1_review.json
Analytics JSON        : G:\site-safety-vision\outputs\predictions\test1_analytics.json
Incident snapshots    : G:\site-safety-vision\outputs\predictions\incidents
--------------------------------------------------------------------------------
(.venv) PS G:\site-safety-vision>
```

Image2:Safety analysis report

```
--------------------------------------------------------------------------------
Processing started...
[Frame 0] workers=8 | safe=7 | unsafe=0 | uncertain=1 | alerts=8
[Frame 30] workers=8 | safe=7 | unsafe=0 | uncertain=1 | alerts=8
[Frame 60] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 90] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 120] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 150] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 180] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 210] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 240] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 270] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 300] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 330] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 360] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 390] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 420] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 450] workers=9 | safe=8 | unsafe=1 | uncertain=0 | alerts=9
[Frame 480] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 510] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 540] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 570] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 600] workers=10 | safe=9 | unsafe=0 | uncertain=1 | alerts=10
[Frame 630] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 660] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 690] workers=9 | safe=9 | unsafe=0 | uncertain=0 | alerts=9
[Frame 720] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 750] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 780] workers=10 | safe=10 | unsafe=0 | uncertain=0 | alerts=10
[Frame 810] workers=9 | safe=8 | unsafe=0 | uncertain=1 | alerts=9
[Frame 840] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 870] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 900] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 930] workers=6 | safe=6 | unsafe=0 | uncertain=0 | alerts=6
[Frame 960] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 990] workers=7 | safe=7 | unsafe=0 | uncertain=0 | alerts=7
[Frame 1020] workers=6 | safe=6 | unsafe=0 | uncertain=0 | alerts=6
[Frame 1050] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 1080] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
[Frame 1110] workers=8 | safe=8 | unsafe=0 | uncertain=0 | alerts=8
--------------------------------------------------------------------------------
```

Image3:Frame by frame analysis